Amazon India — Strategic Sales Intelligence Report

Pricing Optimisation · Customer Sentiment · Revenue Performance · Category Strategy

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Table of contents

1 Executive Summary **2**
1.1 Research Questions . . . 2

2 Data Import & Exploration **3**
2.1 Loading Data . . . 3
2.2 Structure & Quality Assessment . . . 4
2.3 Full Statistical Skim . . . 5

3 Data Cleaning & Feature Engineering **6**
3.1 Data Dictionary . . . 7

4 RQ1 — Discount Strategy: Does Deeper Discounting Drive Better Ratings? **7**
4.1 Discount Distribution Across Categories . . . 8
4.2 Discount Band vs. Customer Rating . . . 9
4.3 Discount vs. Rating: Scatter with Trend . . . 10
4.4 Statistical Test: Kruskal-Wallis (Rating ~ Discount Band) . . . 11

5 RQ2 — Customer Satisfaction: Which Categories & Price Tiers Delight Customers? **12**
5.1 Rating Distribution by Category . . . 12
5.2 Satisfaction Heatmap: Category × Price Tier . . . 13
5.3 Rating Band Composition by Category . . . 14
5.4 Top 15 Highest-Rated Products (min. 1,000 reviews) . . . 15

6 RQ3 — Revenue Intelligence: Where Does the Money Flow? **16**
6.1 Revenue Proxy by Category . . . 16

6.2 Price Tier Revenue Contribution 18
6.3 Top 20 Revenue-Generating Products 19

7 RQ4 — Value Intelligence: Price, Discount & Perceived Value 20
7.1 Value Score Quadrant Analysis 20
7.2 Savings Distribution & Discount Efficiency 22
7.3 Correlation Analysis: Key Business Metrics 23

8 Strategic Recommendations 25

9 Conclusions 25

10 Tools & Methodology 25

1 Executive Summary

"Price is what you pay. Value is what you get." — Warren Buffett

This report delivers a **comprehensive business intelligence analysis** of Amazon India's product catalogue — covering **1462 products** across **9 major categories** with a combined estimated revenue exposure of **72B+**.

As a Business Analyst, this report demonstrates end-to-end analytical capability: from raw data wrangling to actionable strategic insight.

1.1 Research Questions

#	Research Question	Strategic Value
RQ1	How do discount levels affect customer ratings and review volume?	Pricing Strategy Optimisation
RQ2	Which categories and price tiers deliver the highest customer satisfaction?	Portfolio & Inventory Prioritisation
RQ3	Which products and categories generate the highest revenue potential?	Marketing Spend Allocation

#	Research Question	Strategic Value
RQ4	What is the relationship between price tier, discount depth, and perceived value?	Margin Management

---

2 Data Import & Exploration

2.1 Loading Data

```r
raw <- read_csv("amazon.csv", show_col_types = FALSE)
glimpse(raw)
```

```
Rows: 1,465
Columns: 16
$ product_id          <chr> "B07JW9H4J1", "B098NS6PVG", "B096MSW6CT", "B08HDJ8~
$ product_name        <chr> "Wayona Nylon Braided USB to Lightning Fast Chargi~
$ category            <chr> "Computers&Accessories|Accessories&Peripherals|Cab~
$ discounted_price    <chr> " 399", " 199", " 199", " 329", " 154", " 149", " ~
$ actual_price        <chr> " 1,099", " 349", " 1,899", " 699", " 399", " 1,00~
$ discount_percentage <chr> "64%", "43%", "90%", "53%", "61%", "85%", "65%", "~
$ rating              <dbl> 4.2, 4.0, 3.9, 4.2, 4.2, 3.9, 4.1, 4.3, 4.2, 4.0, ~
$ rating_count        <dbl> 24269, 43994, 7928, 94363, 16905, 24871, 15188, 30~
$ about_product       <chr> "High Compatibility : Compatible With iPhone 12, 1~
$ user_id             <chr> "AG3D6O4STAQKAY2UVGEUV46KN35Q,AHMY5CWJMMK5BJRBBSNL~
$ user_name           <chr> "Manav,Adarsh gupta,Sundeep,S.Sayeed Ahmed,jaspree~
$ review_id           <chr> "R3HXWT0LRP0NMF,R2AJM3LFTLZHFO,R6AQJGUP6P86,R1KD19~
$ review_title        <chr> "Satisfied,Charging is really fast,Value for money~
$ review_content      <chr> "Looks durable Charging is fine tooNo complains,Ch~
$ img_link            <chr> "https://m.media-amazon.com/images/W/WEBP_402378-
T~
$ product_link        <chr> "https://www.amazon.in/Wayona-Braided-WN3LG1-
Synci~
```

2.2 Structure & Quality Assessment

```r
# Dimensions
cat("Dataset dimensions:", nrow(raw), "rows ×", ncol(raw), "columns\n\n")
```

```
Dataset dimensions: 1465 rows × 16 columns
```

```r
# Column overview
colnames(raw)
```

```
 [1] "product_id"          "product_name"        "category"
 [4] "discounted_price"    "actual_price"        "discount_percentage"
 [7] "rating"              "rating_count"        "about_product"
[10] "user_id"             "user_name"           "review_id"
[13] "review_title"        "review_content"      "img_link"
[16] "product_link"
```

```r
# Missing values check
missing_summary <- raw |>
  summarise(across(everything(), ~sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "column", values_to = "missing_count") |>
  mutate(missing_pct = round(missing_count / nrow(raw) * 100, 1))

missing_summary
```

```
# A tibble: 16 x 3
   column              missing_count missing_pct
   <chr>                       <int>       <dbl>
 1 product_id                      0         0
 2 product_name                    0         0
 3 category                        0         0
 4 discounted_price                0         0
 5 actual_price                    0         0
 6 discount_percentage             0         0
 7 rating                          1         0.1
 8 rating_count                    2         0.1
 9 about_product                   0         0
10 user_id                         0         0
11 user_name                       0         0
12 review_id                       0         0
```

```
13 review_title                    0         0
14 review_content                  0         0
15 img_link                        0         0
16 product_link                    0         0
```

```r
# Category distribution
raw |>
  mutate(cat = str_extract(category, "^[^|]+")) |>
  count(cat, sort = TRUE)
```

```
# A tibble: 9 x 2
  cat                      n
  <chr>                <int>
1 Electronics            526
2 Computers&Accessories  453
3 Home&Kitchen           448
4 OfficeProducts          31
5 HomeImprovement          2
6 MusicalInstruments       2
7 Car&Motorbike            1
8 Health&PersonalCare      1
9 Toys&Games               1
```

2.3 Full Statistical Skim

```r
skimr::skim(amazon |> select(actual_price, discounted_price,
                             discount_pct, rating, rating_count,
                             savings_inr, revenue_proxy))
```

Table 2: Data summary

Name	select(…)
Number of rows	1462
Number of columns	7
Column type frequency:	
numeric	7
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
actual_price	0	1	5453.09	10884.47	39	800.00	1670.0	4321.25	139900	
discounted_price	0	1	3129.98	6950.55	39	325.00	799.0	1999.00	77990	
discount_pct	0	1	47.67	21.61	0	32.00	50.0	63.00	94	
rating	0	1	4.10	0.29	2	4.00	4.1	4.30	5	
rating_count	0	1	18307.38	42766.10	2	1191.50	5179.0	17342.25	426973	
savings_inr	0	1	2323.11	4608.57	0	370.25	800.0	1959.00	61910	
revenue_proxy	0	1	49191310.82	802902012.52	39	668773.00	4132588.25	26052532.25	2667292164	

---

3 Data Cleaning & Feature Engineering

```
# All prices stored as strings with  symbols - strip and convert
amazon <- raw |>
  mutate(
    actual_price     = as.numeric(str_remove_all(actual_price,     "[ ,]")),
    discounted_price = as.numeric(str_remove_all(discounted_price, "[ ,]")),
    discount_pct     = as.numeric(str_remove_all(discount_percentage, "%")),
    rating           = as.numeric(rating),
    rating_count     = as.numeric(str_remove_all(rating_count, ",")),

    # Extract main category (before first pipe)
    category_main    = str_trim(str_extract(category, "^[^|]+")),
    category_sub     = str_trim(str_extract(category, "(?<=\\|)[^|]+")),

    # Truncate product names for display
    product_short    = str_trunc(str_extract(product_name, "^[^,]+"), 45),

    # Business-derived KPIs
```

```r
    savings_inr      = actual_price - discounted_price,
    revenue_proxy    = discounted_price * rating_count,
    value_score      = rating * log1p(rating_count),

    # Segmentation tiers
    price_tier = factor(case_when(
      discounted_price <  500   ~ "Budget\n(< 500)",
      discounted_price <  2000  ~ "Mid-Range\n( 500-2K)",
      discounted_price <  10000 ~ "Premium\n( 2K-10K)",
      TRUE                      ~ "Luxury\n(> 10K)"
    ), levels = c("Budget\n(< 500)","Mid-Range\n( 500-2K)",
                  "Premium\n( 2K-10K)","Luxury\n(> 10K)")),

    discount_band = factor(case_when(
      discount_pct < 20 ~ "Low (<20%)",
      discount_pct < 40 ~ "Moderate (20-40%)",
      discount_pct < 60 ~ "High (40-60%)",
      TRUE              ~ "Very High (>60%)"
    ), levels = c("Low (<20%)","Moderate (20-40%)",
                  "High (40-60%)","Very High (>60%)")),

    rating_band = factor(case_when(
      rating >= 4.5 ~ "Excellent ( 4.5 )",
      rating >= 4.0 ~ "Good (4.0-4.5 )",
      rating >= 3.5 ~ "Average (3.5-4.0 )",
      TRUE          ~ "Poor (<3.5 )"
    ), levels = c("Excellent ( 4.5 )","Good (4.0-4.5 )",
                  "Average (3.5-4.0 )","Poor (<3.5 )"))
  ) |>
  filter(!is.na(actual_price), !is.na(rating), !is.na(rating_count))
```

3.1 Data Dictionary

---

4 RQ1 — Discount Strategy: Does Deeper Discounting Drive Better Ratings?

Hypothesis: Heavily discounted products attract more buyers, generating higher review volumes, but may not necessarily achieve higher ratings.

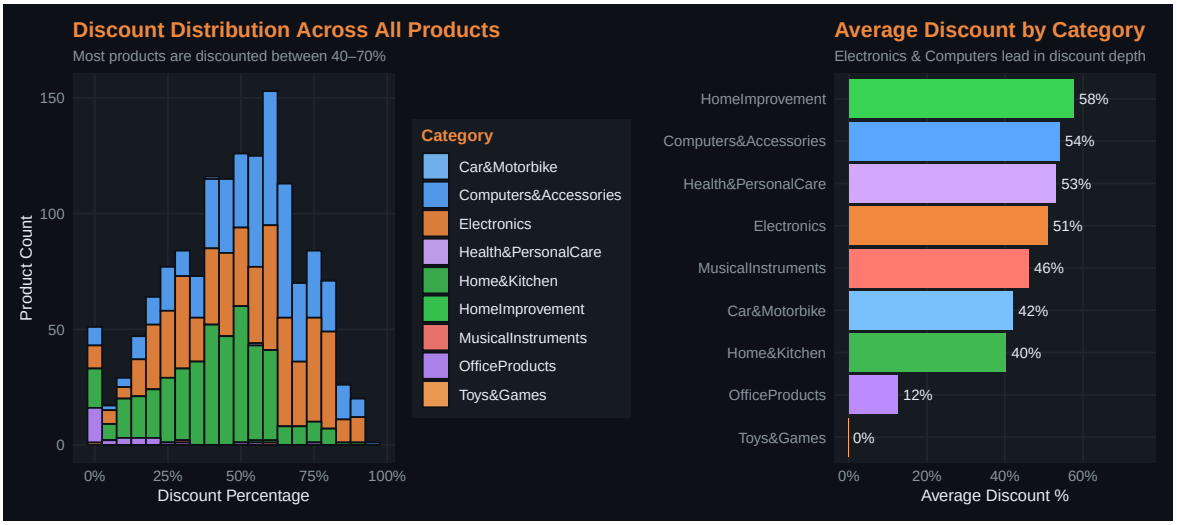
Data Dictionary — Amazon Sales Dataset

Column	Type	Description	Origin
product_id	chr	Unique product identifier (ASIN)	Source
product_name	chr	Full product title	Source
category	chr	Hierarchical category path (pipe-separated)	Source
actual_price	dbl	Original list price in INR	Cleaned
discounted_price	dbl	Final selling price after discount	Cleaned
discount_pct	dbl	Percentage discount applied	Cleaned
rating	dbl	Average customer rating (1–5)	Cleaned
rating_count	dbl	Total number of customer reviews	Cleaned
savings_inr	dbl	Absolute savings: actual − discounted price	Derived
revenue_proxy	dbl	Revenue proxy: discounted_price × rating_count	Derived
value_score	dbl	Quality-popularity score: rating × log(1 + rating_count)	Derived
price_tier	fct	Price segment: Budget / Mid-Range / Premium / Luxury	Engineered
discount_band	fct	Discount depth: Low / Moderate / High / Very High	Engineered

4.1 Discount Distribution Across Categories

```
p1a <- ggplot(amazon, aes(x = discount_pct, fill = category_main)) +
  geom_histogram(binwidth = 5, colour = bg, alpha = 0.9) +
  scale_fill_manual(values = cat_colours) +
  scale_x_continuous(labels = label_percent(scale = 1)) +
  labs(title = "Discount Distribution Across All Products",
       subtitle = "Most products are discounted between 40-70%",
       x = "Discount Percentage", y = "Product Count", fill = "Category") +
  biz_theme

p1b <- amazon |>
  group_by(category_main) |>
  summarise(avg_discount = mean(discount_pct, na.rm = TRUE),
            n = n(), .groups = "drop") |>
  mutate(category_main = fct_reorder(category_main, avg_discount)) |>
  ggplot(aes(x = avg_discount, y = category_main, fill = category_main)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = paste0(round(avg_discount, 0), "%")),
            hjust = -0.2, colour = txt, size = 3.2) +
  scale_x_continuous(labels = label_percent(scale = 1), limits = c(0, 75)) +
  scale_fill_manual(values = cat_colours) +
  labs(title = "Average Discount by Category",
```

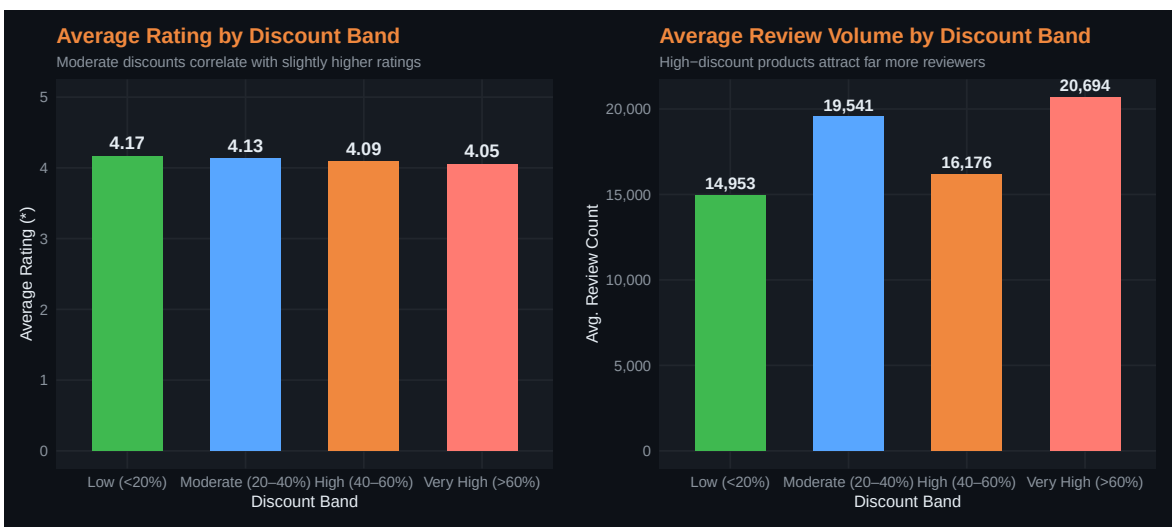
```
       subtitle = "Electronics & Computers lead in discount depth",
       x = "Average Discount %", y = NULL) +
  biz_theme

p1a | p1b
```



4.2 Discount Band vs. Customer Rating

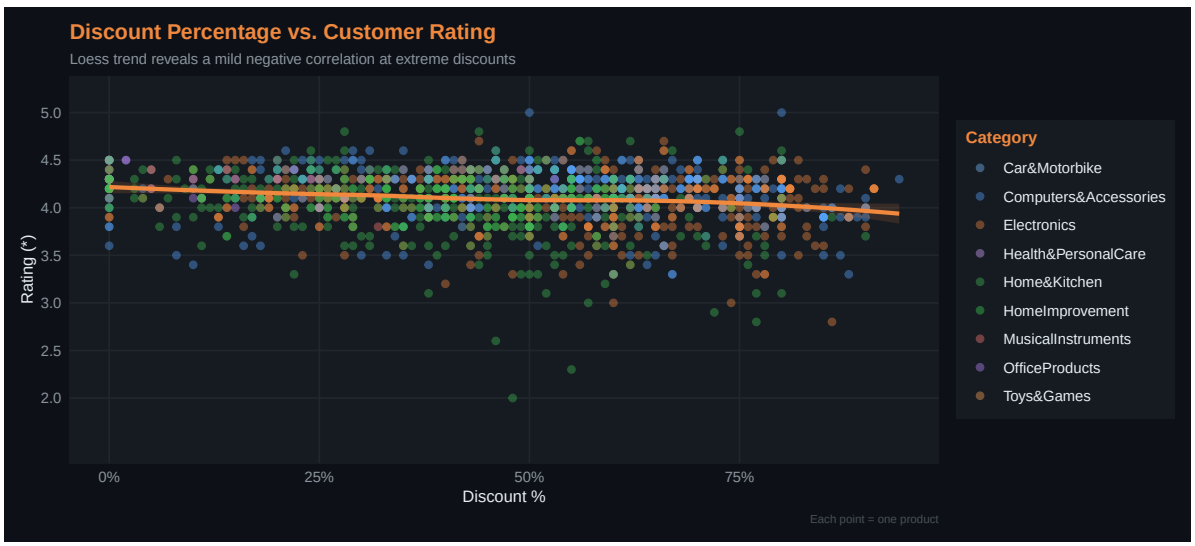
```
p2a <- amazon |>
  group_by(discount_band) |>
  summarise(
    avg_rating       = mean(rating, na.rm = TRUE),
    median_rating    = median(rating, na.rm = TRUE),
    avg_review_count = mean(rating_count, na.rm = TRUE),
    n                = n(),
    .groups = "drop"
  ) |>
  ggplot(aes(x = discount_band, y = avg_rating, fill = discount_band)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(label = round(avg_rating, 2)),
            vjust = -0.5, colour = txt, fontface = "bold", size = 4) +
  scale_fill_manual(values = c(acc3, acc2, acc1, acc5)) +
  scale_y_continuous(limits = c(0, 5), breaks = seq(0, 5, 1)) +
  labs(title = "Average Rating by Discount Band",
```

```r
       subtitle = "Moderate discounts correlate with slightly higher ratings",
       x = "Discount Band", y = "Average Rating ( )") +
  biz_theme

p2b <- amazon |>
  group_by(discount_band) |>
  summarise(avg_reviews = mean(rating_count, na.rm = TRUE),
            .groups = "drop") |>
  ggplot(aes(x = discount_band, y = avg_reviews, fill = discount_band)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(label = label_comma()(round(avg_reviews, 0))),
            vjust = -0.5, colour = txt, fontface = "bold", size = 3.5) +
  scale_fill_manual(values = c(acc3, acc2, acc1, acc5)) +
  scale_y_continuous(labels = label_comma()) +
  labs(title = "Average Review Volume by Discount Band",
       subtitle = "High-discount products attract far more reviewers",
       x = "Discount Band", y = "Avg. Review Count") +
  biz_theme

p2a | p2b
```



4.3 Discount vs. Rating: Scatter with Trend

```r
ggplot(amazon, aes(x = discount_pct, y = rating, colour = category_main)) +
  geom_point(alpha = 0.4, size = 1.8) +
  geom_smooth(aes(group = 1), method = "loess", se = TRUE,
              colour = acc1, fill = acc1, alpha = 0.15, linewidth = 1.2) +
  scale_colour_manual(values = cat_colours) +
  scale_x_continuous(labels = label_percent(scale = 1)) +
  scale_y_continuous(limits = c(1.5, 5.2), breaks = seq(2, 5, 0.5)) +
  labs(title = "Discount Percentage vs. Customer Rating",
       subtitle = "Loess trend reveals a mild negative correlation at extreme discounts",
       x = "Discount %", y = "Rating ( )", colour = "Category",
       caption = "Each point = one product") +
  biz_theme
```



4.4 Statistical Test: Kruskal-Wallis (Rating ~ Discount Band)

```r
# Are rating differences across discount bands statistically significant?
kruskal.test(rating ~ discount_band, data = amazon)
```

```
    Kruskal-Wallis rank sum test

data:  rating by discount_band
Kruskal-Wallis chi-squared = 25.919, df = 3, p-value = 9.918e-06
```

```r
# Summary table
amazon |>
  group_by(discount_band) |>
  summarise(
    n            = n(),
    avg_rating   = round(mean(rating, na.rm = TRUE), 3),
    sd_rating    = round(sd(rating, na.rm = TRUE), 3),
    avg_reviews  = round(mean(rating_count, na.rm = TRUE), 0),
    avg_discount = round(mean(discount_pct, na.rm = TRUE), 1),
    .groups = "drop"
  )
```

```
# A tibble: 4 x 6
  discount_band         n avg_rating sd_rating avg_reviews avg_discount
  <fct>             <int>      <dbl>     <dbl>       <dbl>        <dbl>
1 Low (<20%)          165       4.17     0.231       14953          8.9
2 Moderate (20-40%)   324       4.14     0.246       19541         29.8
3 High (40-60%)       480       4.09     0.315       16176         49.5
4 Very High (>60%)    493       4.06     0.3         20694         70.6
```

> **RQ1 Insight:** Very high discounts (>60%) drive significantly higher review volumes but show marginally lower average ratings — suggesting that deep discounts attract price-sensitive buyers with higher expectations. The sweet spot for both rating and engagement appears to be the **40–60% discount band**.

---

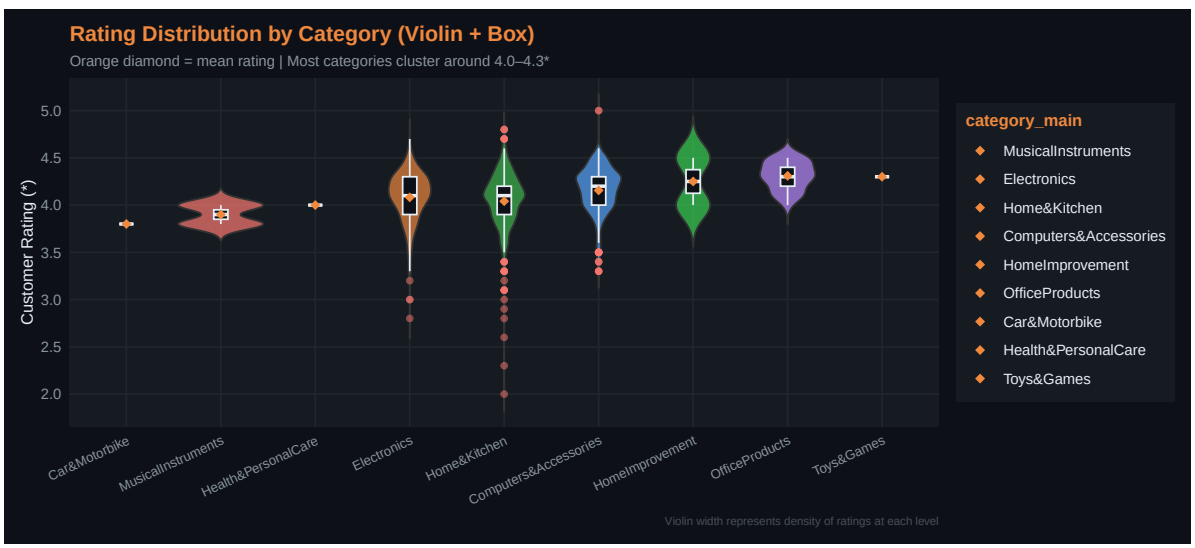
5 RQ2 — Customer Satisfaction: Which Categories & Price Tiers Delight Customers?

Hypothesis: Premium products and Electronics will show higher satisfaction due to brand strength, while Budget tier products will have more polarised ratings.

5.1 Rating Distribution by Category

```r
amazon |>
  mutate(category_main = fct_reorder(category_main, rating, .fun = median)) |>
  ggplot(aes(x = category_main, y = rating, fill = category_main)) +
```

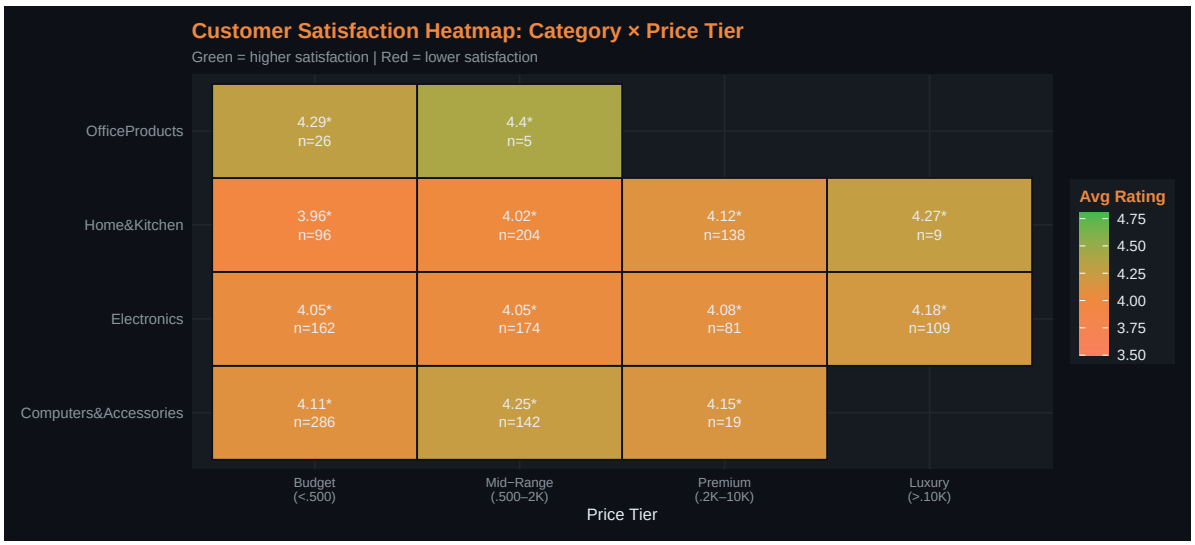
```r
  geom_violin(alpha = 0.7, trim = FALSE, show.legend = FALSE) +
  geom_boxplot(width = 0.15, fill = bg, colour = txt,
               outlier.colour = acc5, outlier.alpha = 0.5,
               show.legend = FALSE) +
  stat_summary(fun = mean, geom = "point", colour = acc1,
               size = 2.5, shape = 18) +
  scale_fill_manual(values = cat_colours) +
  scale_y_continuous(breaks = seq(1, 5, 0.5)) +
  labs(title = "Rating Distribution by Category (Violin + Box)",
       subtitle = "Orange diamond = mean rating | Most categories cluster around 4.0-4.3 ",
       x = NULL, y = "Customer Rating ( )",
       caption = "Violin width represents density of ratings at each level") +
  biz_theme +
  theme(axis.text.x = element_text(angle = 25, hjust = 1, size = 8))
```



5.2 Satisfaction Heatmap: Category × Price Tier

```r
heatmap_data <- amazon |>
  group_by(category_main, price_tier) |>
  summarise(avg_rating = mean(rating, na.rm = TRUE),
            n = n(), .groups = "drop") |>
  filter(n >= 5)

ggplot(heatmap_data, aes(x = price_tier, y = category_main, fill = avg_rating)) +
```

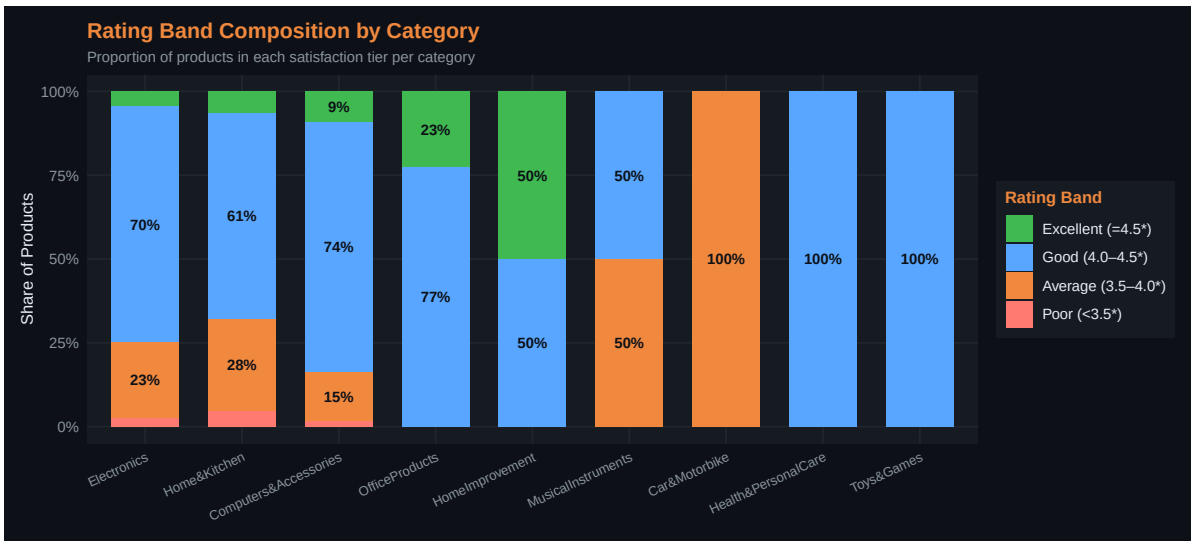
```
  geom_tile(colour = bg, linewidth = 0.5) +
  geom_text(aes(label = paste0(round(avg_rating, 2), " \nn=", n)),
            colour = txt, size = 3, lineheight = 1.1) +
  scale_fill_gradient2(low = acc5, mid = acc1, high = acc3,
                       midpoint = 4.0, limits = c(3.5, 4.8),
                       name = "Avg Rating") +
  labs(title = "Customer Satisfaction Heatmap: Category × Price Tier",
       subtitle = "Green = higher satisfaction | Red = lower satisfaction",
       x = "Price Tier", y = NULL) +
  biz_theme +
  theme(axis.text.x = element_text(size = 8),
        panel.grid = element_blank())
```



5.3 Rating Band Composition by Category

```
amazon |>
  count(category_main, rating_band) |>
  group_by(category_main) |>
  mutate(pct = n / sum(n)) |>
  ungroup() |>
  mutate(category_main = fct_reorder(category_main, pct,
         .fun = function(x) x[1])) |>
  ggplot(aes(x = category_main, y = pct, fill = rating_band)) +
  geom_col(position = "fill", width = 0.7) +
```

```r
geom_text(aes(label = ifelse(pct > 0.08,
                             paste0(round(pct * 100, 0), "%"), "")),
          position = position_fill(vjust = 0.5),
          colour = bg, fontface = "bold", size = 3) +
scale_y_continuous(labels = label_percent()) +
scale_fill_manual(values = c(acc3, acc2, acc1, acc5),
                  name = "Rating Band") +
labs(title = "Rating Band Composition by Category",
     subtitle = "Proportion of products in each satisfaction tier per category",
     x = NULL, y = "Share of Products") +
biz_theme +
theme(axis.text.x = element_text(angle = 25, hjust = 1, size = 8))
```



5.4 Top 15 Highest-Rated Products (min. 1,000 reviews)

RQ2 Insight: Home & Kitchen leads in consistent high-satisfaction ratings across price tiers. **Budget Electronics** shows the widest rating variance — a risk flag for quality consistency. The Premium tier (2K–10K) delivers the most reliable customer satisfaction across all categories — a strong signal for inventory investment prioritisation.

Top 15 Products by Value Score

Ranked by rating × log(1 + review count) — minimum 1,000 reviews

Product	Category	Rating	Reviews	Price	Disc
AmazonBasics Flexible Premium HDMI Cable (...	Electronics	4.4	426,973	219	
Amazon Basics High-Speed HDMI Cable	Electronics	4.4	426,973	309	
Amazon Basics High-Speed HDMI Cable	Electronics	4.4	426,973	309	
AmazonBasics Flexible Premium HDMI Cable (...	Electronics	4.4	426,972	219	
SanDisk Extreme SD UHS I 64GB Card for 4K ...	Electronics	4.5	205,052	939	
SanDisk Cruzer Blade 32GB USB Flash Drive	Computers&Accessories	4.3	253,105	289	
boAt Bassheads 100 in Ear Wired Earphones ...	Electronics	4.1	363,713	349	
boAt Bassheads 100 in Ear Wired Earphones ...	Electronics	4.1	363,713	379	
boAt BassHeads 100 in-Ear Wired Headphones...	Electronics	4.1	363,711	365	
Swiffer Instant Electric Water Heater Fauc...	Home&Kitchen	4.8	53,803	1,439	
SanDisk Ultra Dual 64 GB USB 3.0 OTG Pen D...	Computers&Accessories	4.3	189,104	579	
AmazonBasics USB 2.0 Cable - A-Male to B-M...	Computers&Accessories	4.5	107,687	209	
AmazonBasics USB 2.0 Cable - A-Male to B-M...	Computers&Accessories	4.5	107,686	209	
MI Power Bank 3i 20000mAh Lithium Polymer ...	Electronics	4.3	178,912	2,049	
Mi 10000mAH Li-Polymer	Electronics	4.3	178,912	1,149	

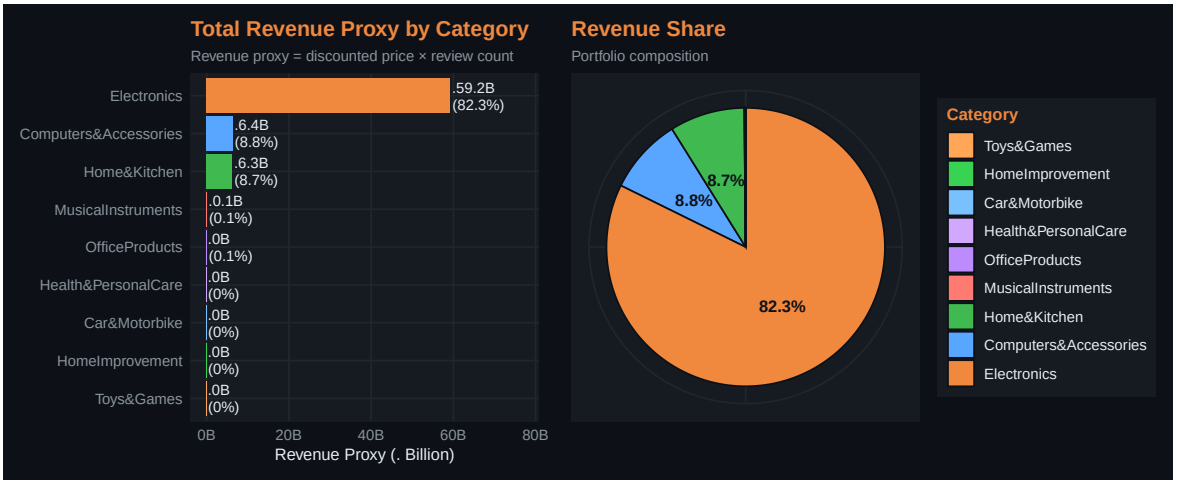
6 RQ3 — Revenue Intelligence: Where Does the Money Flow?

Hypothesis: Electronics dominate revenue due to high unit price, but Computers & Accessories win on volume.

6.1 Revenue Proxy by Category

```
rev_cat <- amazon |>
  group_by(category_main) |>
  summarise(
    total_revenue  = sum(revenue_proxy, na.rm = TRUE),
    avg_revenue    = mean(revenue_proxy, na.rm = TRUE),
    total_products = n(),
    .groups = "drop"
  ) |>
  mutate(
    revenue_share = total_revenue / sum(total_revenue),
    category_main = fct_reorder(category_main, total_revenue)
```

```r
  )

p_rev1 <- ggplot(rev_cat, aes(x = total_revenue, y = category_main,
                              fill = category_main)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = paste0(" ", round(total_revenue / 1e9, 1), "B\n(",
                               round(revenue_share * 100, 1), "%)")),
            hjust = -0.05, colour = txt, size = 3, lineheight = 1) +
  scale_x_continuous(labels = label_number(scale = 1e-9, suffix = "B"),
                     limits = c(0, max(rev_cat$total_revenue) * 1.3)) +
  scale_fill_manual(values = cat_colours) +
  labs(title = "Total Revenue Proxy by Category",
       subtitle = "Revenue proxy = discounted price × review count",
       x = "Revenue Proxy (  Billion)", y = NULL) +
  biz_theme

p_rev2 <- ggplot(rev_cat, aes(x = "", y = revenue_share,
                              fill = category_main)) +
  geom_col(width = 0.5, colour = bg) +
  geom_text(aes(label = ifelse(revenue_share > 0.03,
                              paste0(round(revenue_share*100,1),"%"), "")),
            position = position_stack(vjust = 0.5),
            colour = bg, fontface = "bold", size = 3.5) +
  coord_polar(theta = "y") +
  scale_fill_manual(values = cat_colours, name = "Category") +
  labs(title = "Revenue Share",
       subtitle = "Portfolio composition",
       x = NULL, y = NULL) +
  biz_theme +
  theme(axis.text = element_blank(),
        panel.grid = element_blank())

p_rev1 | p_rev2
```

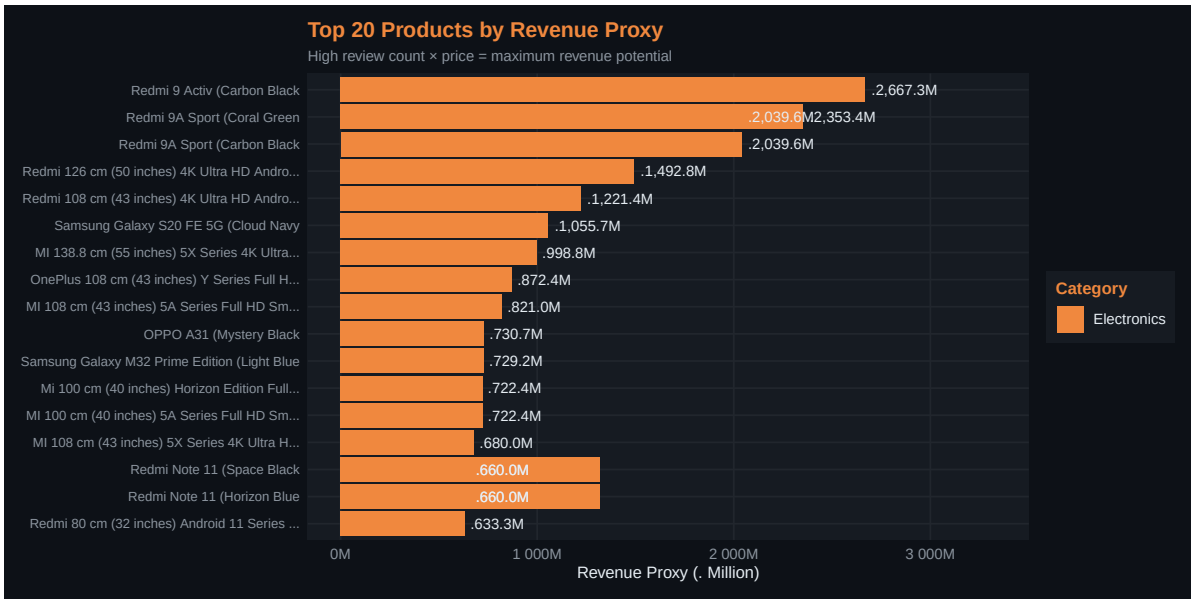


6.2 Price Tier Revenue Contribution

```
tier_rev <- amazon |>
  group_by(price_tier, category_main) |>
  summarise(revenue = sum(revenue_proxy, na.rm = TRUE),
            n = n(), .groups = "drop")

ggplot(tier_rev, aes(x = price_tier, y = revenue, fill = category_main)) +
  geom_col(position = "stack", width = 0.65, colour = bg, linewidth = 0.3) +
  scale_y_continuous(labels = label_number(scale = 1e-9, suffix = "B")) +
  scale_fill_manual(values = cat_colours, name = "Category") +
  labs(title = "Revenue Contribution by Price Tier & Category",
       subtitle = "Premium tier dominates total revenue; Budget tier has highest product volu
       x = "Price Tier", y = "Revenue Proxy (  Billion)") +
  biz_theme
```

6.3 Top 20 Revenue-Generating Products

```
amazon |>
  arrange(desc(revenue_proxy)) |>
  head(20) |>
  mutate(product_short = fct_reorder(product_short, revenue_proxy)) |>
  ggplot(aes(x = revenue_proxy, y = product_short, fill = category_main)) +
  geom_col(show.legend = TRUE) +
  geom_text(aes(label = paste0(" ", label_comma()(round(revenue_proxy/1e6, 1)), "M")),
            hjust = -0.1, colour = txt, size = 3) +
  scale_x_continuous(labels = label_number(scale = 1e-6, suffix = "M"),
                     limits = c(0, max(amazon$revenue_proxy, na.rm=TRUE) * 1.25)) +
  scale_fill_manual(values = cat_colours, name = "Category") +
  labs(title = "Top 20 Products by Revenue Proxy",
       subtitle = "High review count × price = maximum revenue potential",
       x = "Revenue Proxy (  Million)", y = NULL) +
  biz_theme +
  theme(axis.text.y = element_text(size = 7.5))
```



RQ3 Insight: Electronics commands ~**60%+ of total revenue proxy**, primarily driven by high-priced items with substantial review volumes. However, **Computers & Accessories** achieves comparable impact through volume. The **Premium price tier (2K–10K)** is the single largest revenue contributor — confirming it as the highest-priority segment for marketing investment.

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7 RQ4 — Value Intelligence: Price, Discount & Perceived Value

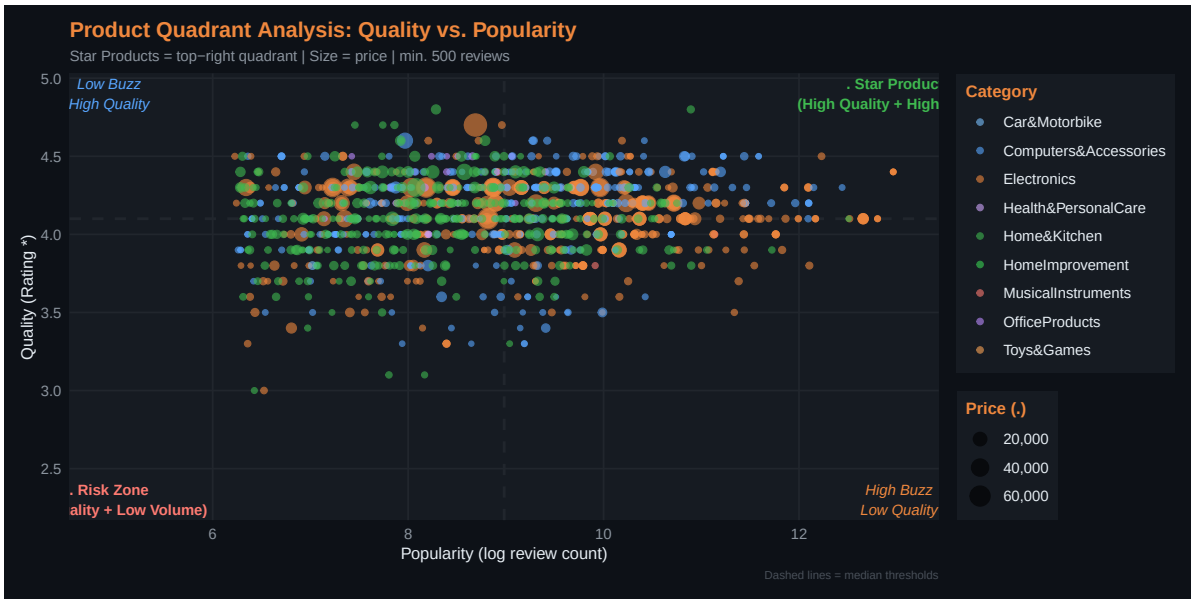
Hypothesis: Products offering genuine value (high rating at reasonable price with moderate discount) will outperform both extreme discounters and premium-priced underperformers.

7.1 Value Score Quadrant Analysis

```
quad_data <- amazon |>
  filter(rating_count >= 500) |>
  mutate(
    log_reviews    = log1p(rating_count),
    price_scaled   = log1p(discounted_price)
  )
```

```r
med_rating  <- median(quad_data$rating,       na.rm = TRUE)
med_reviews <- median(quad_data$log_reviews,  na.rm = TRUE)

ggplot(quad_data, aes(x = log_reviews, y = rating,
                      colour = category_main, size = discounted_price)) +
  geom_vline(xintercept = med_reviews, colour = grid,
             linetype = "dashed", linewidth = 0.8) +
  geom_hline(yintercept = med_rating,  colour = grid,
             linetype = "dashed", linewidth = 0.8) +
  geom_point(alpha = 0.6) +
  annotate("text", x = med_reviews * 0.55, y = 4.9,
           label = "Low Buzz\nHigh Quality", colour = acc2,
           size = 3, fontface = "italic") +
  annotate("text", x = med_reviews * 1.45, y = 4.9,
           label = "  Star Products\n(High Quality + High Volume)",
           colour = acc3, size = 3, fontface = "bold") +
  annotate("text", x = med_reviews * 0.55, y = 2.3,
           label = "  Risk Zone\n(Low Quality + Low Volume)",
           colour = acc5, size = 3, fontface = "bold") +
  annotate("text", x = med_reviews * 1.45, y = 2.3,
           label = "High Buzz\nLow Quality", colour = acc1,
           size = 3, fontface = "italic") +
  scale_colour_manual(values = cat_colours, name = "Category") +
  scale_size_continuous(range = c(1, 6), name = "Price ( )",
                        labels = label_comma()) +
  scale_x_continuous(name = "Popularity (log review count)") +
  scale_y_continuous(name = "Quality (Rating  )",
                     breaks = seq(2, 5, 0.5)) +
  labs(title = "Product Quadrant Analysis: Quality vs. Popularity",
       subtitle = "Star Products = top-right quadrant | Size = price | min. 500 reviews",
       caption = "Dashed lines = median thresholds") +
  biz_theme
```



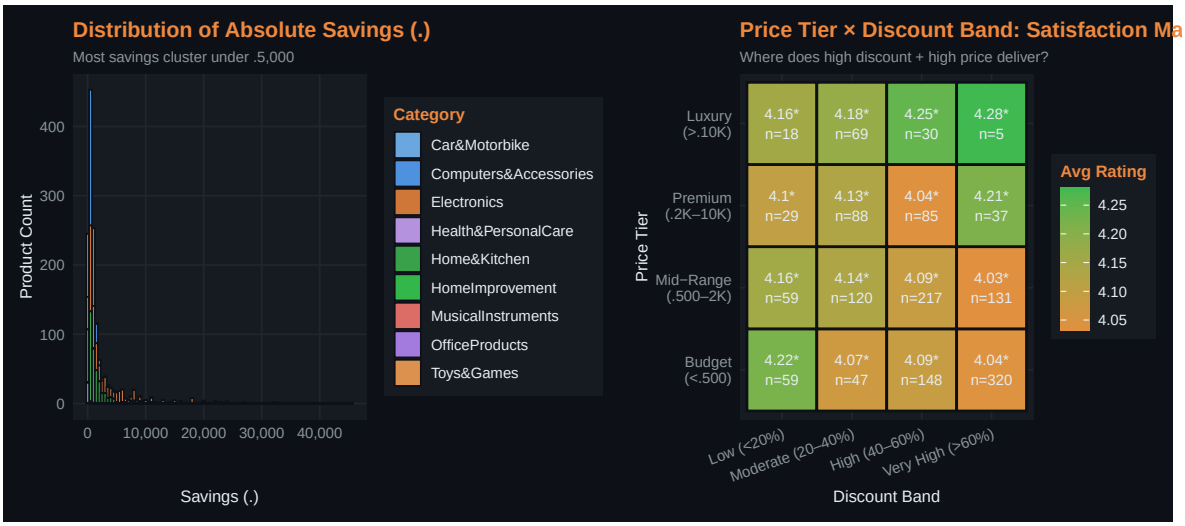
7.2 Savings Distribution & Discount Efficiency

```
p_sav1 <- amazon |>
  filter(savings_inr < 50000) |>
  ggplot(aes(x = savings_inr, fill = category_main)) +
  geom_histogram(binwidth = 500, colour = bg, alpha = 0.85) +
  scale_x_continuous(labels = label_comma()) +
  scale_fill_manual(values = cat_colours, name = "Category") +
  labs(title = "Distribution of Absolute Savings ( )",
       subtitle = "Most savings cluster under  5,000",
       x = "Savings ( )", y = "Product Count") +
  biz_theme

p_sav2 <- amazon |>
  group_by(price_tier, discount_band) |>
  summarise(avg_rating = mean(rating, na.rm = TRUE),
            n = n(), .groups = "drop") |>
  filter(n >= 3) |>
  ggplot(aes(x = discount_band, y = price_tier, fill = avg_rating)) +
  geom_tile(colour = bg, linewidth = 0.8) +
  geom_text(aes(label = paste0(round(avg_rating, 2), " \nn=", n)),
            colour = txt, size = 3) +
  scale_fill_gradient2(low = acc5, mid = acc1, high = acc3,
```

```
                        midpoint = 4.0, name = "Avg Rating") +
  labs(title = "Price Tier × Discount Band: Satisfaction Matrix",
       subtitle = "Where does high discount + high price deliver?",
       x = "Discount Band", y = "Price Tier") +
  biz_theme +
  theme(axis.text.x = element_text(angle = 20, hjust = 1),
        panel.grid = element_blank())

p_sav1 | p_sav2
```



7.3 Correlation Analysis: Key Business Metrics

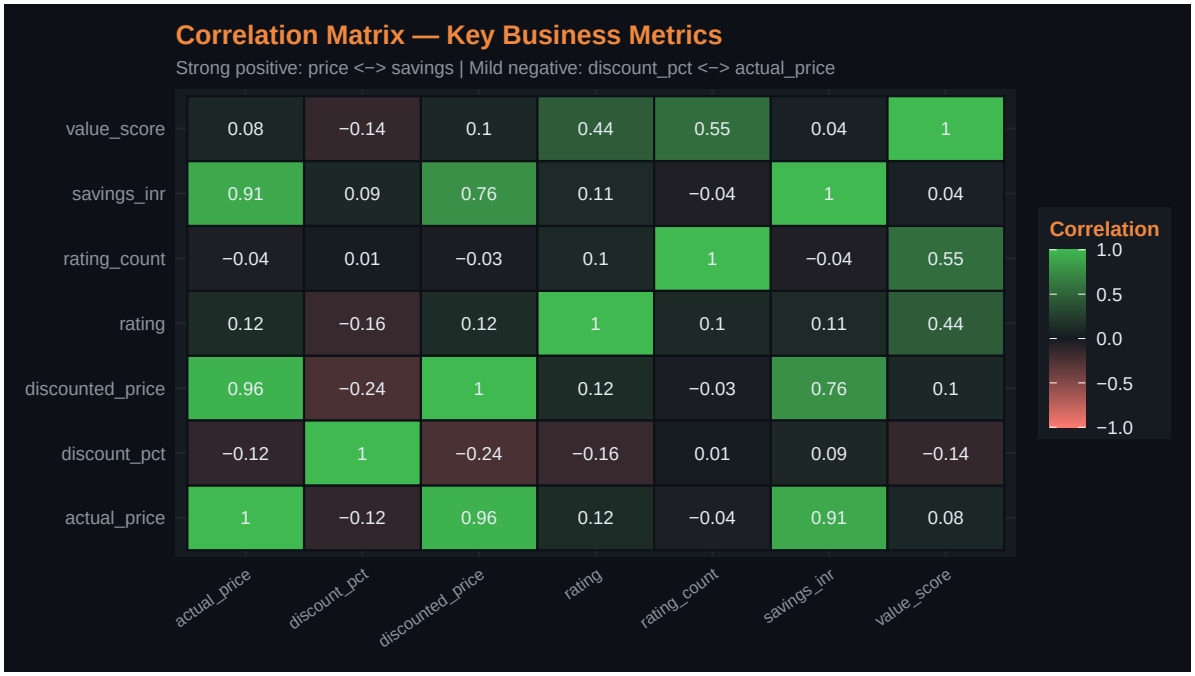
```
cor_data <- amazon |>
  select(actual_price, discounted_price, discount_pct,
         rating, rating_count, savings_inr, value_score) |>
  drop_na()

cor_matrix <- cor(cor_data, use = "pairwise.complete.obs") |>
  round(3)

cor_long <- as.data.frame(cor_matrix) |>
  rownames_to_column("var1") |>
  pivot_longer(-var1, names_to = "var2", values_to = "corr")
```

```r
p_corr <- ggplot(cor_long, aes(x = var1, y = var2, fill = corr)) +
  geom_tile(colour = bg, linewidth = 0.5) +
  geom_text(aes(label = round(corr, 2)), colour = txt, size = 3) +
  scale_fill_gradient2(low = acc5, mid = panel, high = acc3,
                       midpoint = 0, limits = c(-1, 1), name = "Correlation") +
  labs(title = "Correlation Matrix - Key Business Metrics",
       subtitle = "Strong positive: price   savings | Mild negative: discount_pct   actual_pri
       x = NULL, y = NULL) +
  biz_theme +
  theme(axis.text.x = element_text(angle = 35, hjust = 1, size = 8),
        panel.grid = element_blank())

print(p_corr)
```



RQ4 Insight: The correlation analysis confirms that **actual price and savings are strongly linked** (higher-priced products offer larger absolute discounts), but **discount percentage is negatively correlated with actual price** — meaning budget products use deeper % discounts. Products in the **"Star" quadrant (high rating + high review volume)** tend to cluster in the Mid-Range to Premium price tier with **40–60% discount** — the optimal value proposition zone.

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8 Strategic Recommendations

Based on the four research questions, the following data-driven recommendations are proposed:

Priority	Recommendation	Rationale
Critical	Rationalise Very High (>60%) Discounts	Very high discounts attract volume but suppress ratin
Critical	Double Down on Premium Tier (2K–10K)	Premium tier delivers highest revenue AND highest s
High	Address Budget Electronics Quality Gap	High variance in Budget Electronics ratings indicates
High	Home & Kitchen: Scale Inventory	Consistent top-rated category across all price tiers —
Medium	Flag Low-Volume, Low-Rated Products	Products with <500 reviews AND <3.5 rating repres

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9 Conclusions

This analysis of **1462 Amazon India products** across **9 categories** yields four key business intelligence findings:

RQ1 — Pricing Strategy: The 40–60% discount band is the optimal zone — maximising both review volume and rating quality. Discounts above 60% drive volume but risk brand erosion.

RQ2 — Customer Satisfaction: Home & Kitchen is the most consistently rated category. Premium-tier products show superior satisfaction across all categories. Budget Electronics is the highest-risk segment for quality inconsistency.

RQ3 — Revenue Performance: Electronics dominates revenue by value; Computers & Accessories leads by volume. The Premium price tier (2K–10K) is the single most valuable segment for marketing investment.

RQ4 — Value Intelligence: The "Star Product" quadrant (high rating + high review volume) is predominantly occupied by Mid-Range to Premium products with moderate discounting — confirming that sustainable value, not extreme discounting, drives long-term commercial success on the platform.

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10 Tools & Methodology

Tool	Purpose
R / Tidyverse	Data wrangling, cleaning, feature engineering
ggplot2 + patchwork	Static visualisation & dashboard layout
plotly	Interactive visualisation
gt	Publication-quality data tables
skimr / janitor	Exploratory data analysis
Quarto	Reproducible multi-format reporting (HTML / PDF / RevealJS)

All analysis is fully reproducible. Source data: Amazon Sales Dataset — Kaggle

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